Supplementary Guide to Source Code and Model Outputs

This document serves as a guide to understanding the structure of the provided source code, the inference strategy, and the generated outputs evaluated in this study.

1. Notebook Structure and Kaggle Constraints

The complete implementation of the project is provided across two primary Jupyter notebooks: Group21_Source_Code5 and Group21_Source_Code6.

- **Reason for Split:** Training the proposed Multi-Layer Transformer architecture on the Kaggle GPU environment required over 3 hours of compute time. Due to Kaggle's session time limits and memory constraints, the environment session unexpectedly disconnected **("snapped")** during the final image generation phase.
- **Output Location:** As a result of this session interruption, the complete set of generated image outputs for the Multi-Layer model is distributed between these two notebooks. Both notebooks must be reviewed to see the full pipeline and all generated captions.

2. Inference and Beam Search Strategy

During the evaluation phase, captions were generated using Beam Search decoding.

- Because different Beam Widths (K) fundamentally alter the search path, each K value produces a uniquely structured caption—some exhibiting higher semantic accuracy than others.
- To comprehensively evaluate decoding performance, the model was directed to run inference on the exact same set of images **four separate times**, applying Beam Widths of K=1,K=2,K=3, and K=4 respectively. This allowed us to observe how beam expansion affects contextual accuracy.

3. Out-of-Distribution Evaluation Datasets

To rigorously test the model's zero-shot generalization on real-world, out-of-distribution data, two distinct sets of external images were processed:

- **Set 1 (Urban Landscapes):** Comprises 10 images of standard geographic and municipal features.
- **Set 2 (Conflict Zones - 2026 Iran War):** Comprises 28 complex images depicting real-world military and conflict scenarios.

4. Curation for Final Deliverables

Due to the large volume of captions generated across the 38 total images and 4 different Beam Widths, a curation process was applied. The most visually comprehensible images and highly representative captions (highlighting both the successes of the Multi-Layer model and the failures of the baseline) were selected and featured in the final Project Report, Results PDF, and Presentation. The complete, uncurated generation logs remain available in the source notebooks.

5. Directory of Code Notebooks and Output PDFs

To facilitate easy review, the project files are organized numerically. For each phase of the project, the .ipynb files contain the raw source code and training logic, while the corresponding _with_Output_Results.pdf files contain the fully executed notebooks, including the training logs and generated captions.

Phase 1: Baseline Architecture Selection (Flickr8K Dataset)

- **Code 1 (Group21_Source_Code1):** Contains the code and outputs for the **CNN-LSTM** model.
- **Code 2 (Group21_Source_Code2):** Contains the code and outputs for the **CNN-GRU** model.
- **Code 3 (Group21_Source_Code3):** Contains the code and outputs for the **Single Layer Transformer** model.

Phase 2: Remote Sensing Implementation (Satellite Image Dataset)

- **Code 4 (Group21_Source_Code4):** Contains the code and outputs for the **Single Layer Transformer** (Baseline) trained on the satellite imagery.
- **Code 5 & Code 6 (Group21_Source_Code5 & Group21_Source_Code6):** Contain the code and distributed generation outputs for the proposed **Multi-Layer Transformer**. As noted above, Code 5 contains the training process and initial outputs, while Code 6 contains the remainder of the generated captions following the session interruption.